Ansys Zemax OpticStudio 2025 R1

Ansys
Ansys, Inc.
Southpointe
2600 ANSYS Drive
Canonsburg, PA 15317
ansysinfo@ansys.com
(T) 724-746-3304
(F) 724-514-9494

Release 2025 R1
January 2025

Ansys, Inc. and ANSYS Europe, Ltd. are UL registered ISO 9001:2015 companies

Contents

1: The File Tab....30
1.1. Lens File Group....31
1.1.1. New/Open/Save/Save As....31
1.1.2. Insert Lens....31
1.1.3. File Comparator....32
1.2. Archive Group....33
1.2.1. Create Archive....34
1.2.2. Load Archive....35
1.3. Project Directory Group....38
1.3.1. New Lens Project / Save Lens Project As....38
1.3.2. Convert to Lens Project....39
1.3.3. Convert to Lens File....41
1.3.4. Lens Project Settings....42
1.4. Export Group....45
1.4.1. CAD Files....45
1.4.2. Point Cloud....49
1.4.3. DXF/IGES Linework....50
1.4.3.1. Export 2D DXF....50
1.4.3.2. Export 3D DXF....51
1.4.3.3. Export IGES Linework....52
1.4.4. Zemax Black Box....52
1.4.5. Encrypted Coating....55
1.4.6. Export Reduced Order Model to Speos....56
1.4.7. Export Optical Design to Speos....60
1.4.8. Export to PanDao....62
1.5. Convert Group....66
1.5.1. Convert to NSC Group....66
1.5.1.1. Converting sequential surfaces to non-sequential objects....68
1.5.1.2. Exceptions and Restrictions....69
1.5.2. Prepare For OpticsBuilder....75
1.5.2.1. Steps for conversion from Sequential Mode....76
1.5.2.2. Steps for conversion from Non-Sequential Mode....77
1.5.2.3. User Inputs....77
1.5.2.4. Drawing Inputs....78
1.5.2.5. Save .ZBD File....78
1.5.2.6. Conversion errors....79
1.5.3. Convert File Formats....80
1.5.3.1. INT Zernike to OpticStudio DAT....81
1.5.3.2. INT Grid to OpticStudio DAT....82
1.5.3.3. INT Grid to OpticStudio GRD....83

Release 2025 R1 - © Ansys, Inc. All rights reserved.
Contains proprietary and confidential information of ANSYS, Inc. and its subsidiaries and affiliates.

1.5.3.4. Bitmap Image to OpticStudio DAT....85
1.5.3.5. OptiWave® F3d Beam File to OpticStudio ZBF....86
1.5.3.6. Convert ZRD or ZBF to TSV....87
1.5.3.7. Convert CODE V to OpticStudio....87
1.6. Explode Group....95
1.6.1. Explode Autodesk Inventor Assembly....96
1.6.2. Explode Creo Parametric Assembly....96
1.6.3. Explode CAD Assembly....96
1.7. Exit Button....97
2: The Setup Tab....98
2.1. System Group (the Setup Tab)....98
2.1.1. System Explorer....99
2.1.1.1. Aperture (System Explorer)....100
2.1.1.2. Fields....106
2.1.1.3. Wavelengths....120
2.1.1.4. Environment....121
2.1.1.5. Polarization (System Explorer)....129
2.1.1.6. Advanced Options (System Explorer)....145
2.1.1.7. Ray Aiming....151
2.1.1.8. Material Catalogs....157
2.1.1.9. Non-sequential (system explorer)....157
2.1.1.10. Named Filters....161
2.1.1.11. Title/Notes....162
2.1.1.12. Files....162
2.1.1.13. Units....163
2.1.1.14. Cost Estimator (system explorer)....166
2.1.2. OpticStudio Preferences....167
2.1.2.1. Address (OpticStudio preferences)....167
2.1.2.2. Colors....169
2.1.2.3. Editors....170
2.1.2.4. Folders....171
2.1.2.5. General....176
2.1.2.6. Graphics (OpticStudio preferences)....179
2.1.2.7. Toolbar....181
2.1.2.8. Shortcut Keys....183
2.1.2.9. Message Boxes....188
2.1.2.10. Privacy....189
2.1.3. Scale Lens....190
2.1.4. Autosave....192
2.2. Program Mode Group....192
2.2.1. Sequential UI Mode....193
2.2.2. Non-sequential UI Mode....194
2.2.3. Switching UI Modes....194
2.3. Editors Group (Setup Tab)....198